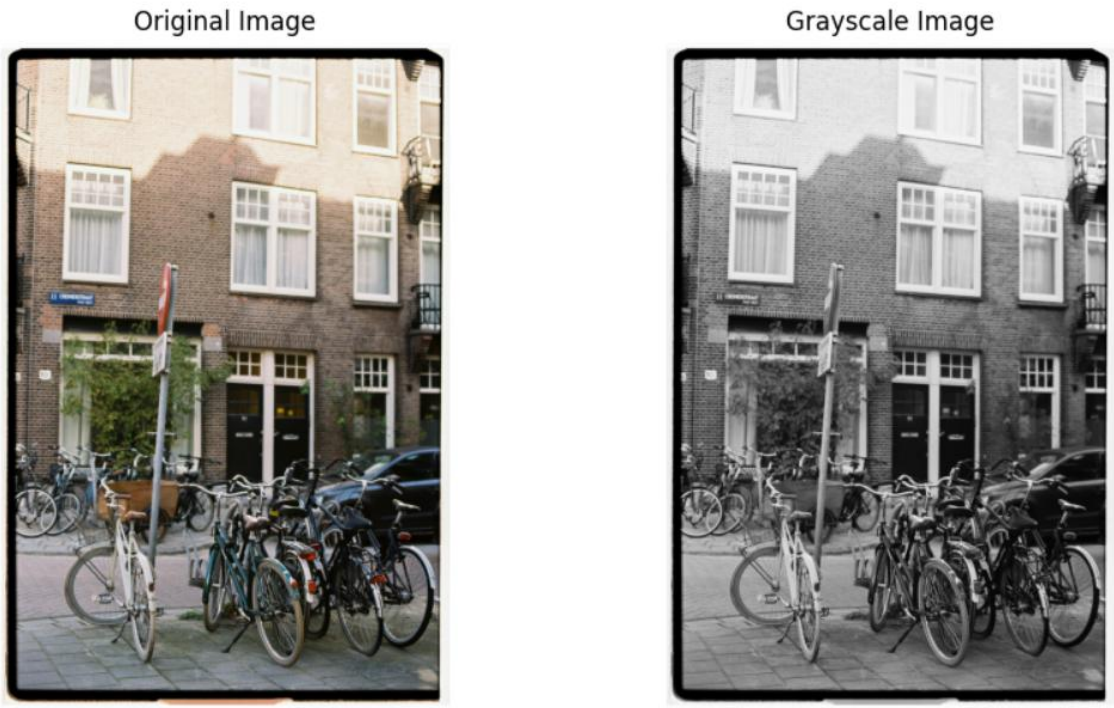
Assignment Tasks

Task 1: Image Selection and Preprocessing

```
import cv2
import matplotlib.pyplot as plt



image = cv2.imread('doc.jpeg')


if image is None:
    print("Error: Image not found.")
    exit()


image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)


gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


plt.figure(figsize=(10,5))


plt.subplot(1,2,1)
plt.title("Original Image")
plt.imshow(image_rgb)
plt.axis('off')


plt.subplot(1,2,2)
plt.title("Grayscale Image")
```

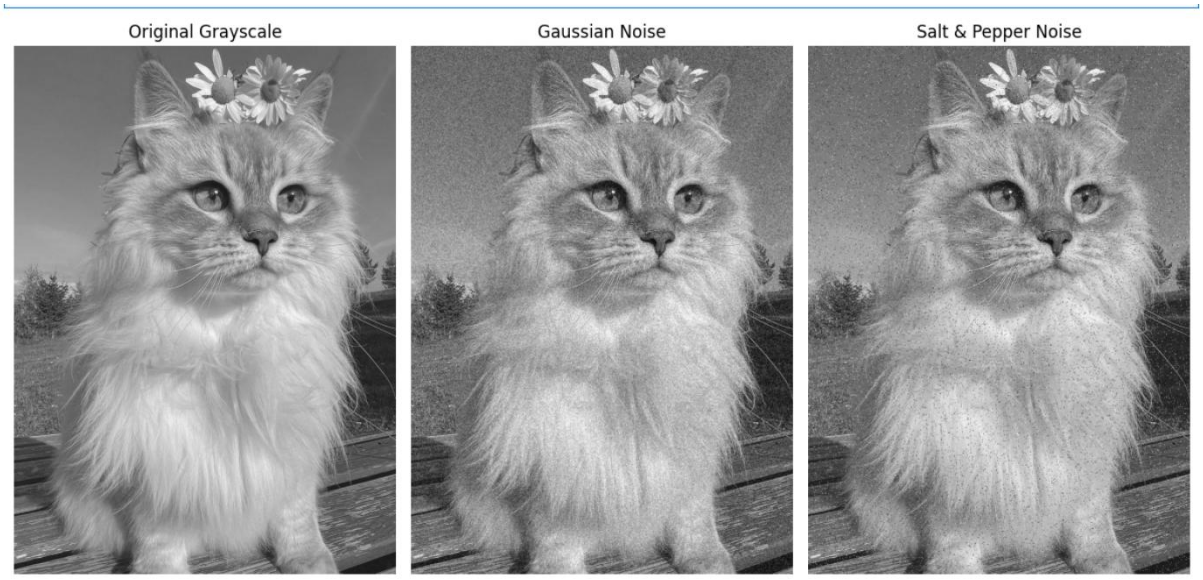
```
plt.imshow(gray_image, cmap='gray')

plt.axis('off')


plt.tight_layout()

plt.show()
```



Task 2: Noise Modeling

```
import numpy as np
import cv2
import matplotlib.pyplot as plt

def add_gaussian_noise(image, mean=0, sigma=25):
    gaussian = np.random.normal(mean, sigma, image.shape)
    noisy_image = image + gaussian
    noisy_image = np.clip(noisy_image, 0, 255)
    return noisy_image.astype(np.uint8)

def add_salt_pepper_noise(image, salt_prob=0.02, pepper_prob=0.02):
    noisy_image = np.copy(image)
    salt = np.random.rand(*image.shape) < salt_prob
    noisy_image[salt] = 255
    pepper = np.random.rand(*image.shape) < pepper_prob
    noisy_image[pepper] = 0
    return noisy_image

gaussian_noisy = add_gaussian_noise(gray_image)
sp_noisy = add_salt_pepper_noise(gray_image)

plt.figure(figsize=(12,6))
```

```
plt.subplot(1,3,1)
plt.title("Original Grayscale")
plt.imshow(gray_image, cmap='gray')
plt.axis('off')

plt.subplot(1,3,2)
plt.title("Gaussian Noise")
plt.imshow(gaussian_noisy, cmap='gray')
plt.axis('off')

plt.subplot(1,3,3)
plt.title("Salt & Pepper Noise")
plt.imshow(sp_noisy, cmap='gray')
plt.axis('off')

plt.tight_layout()
plt.show()
```

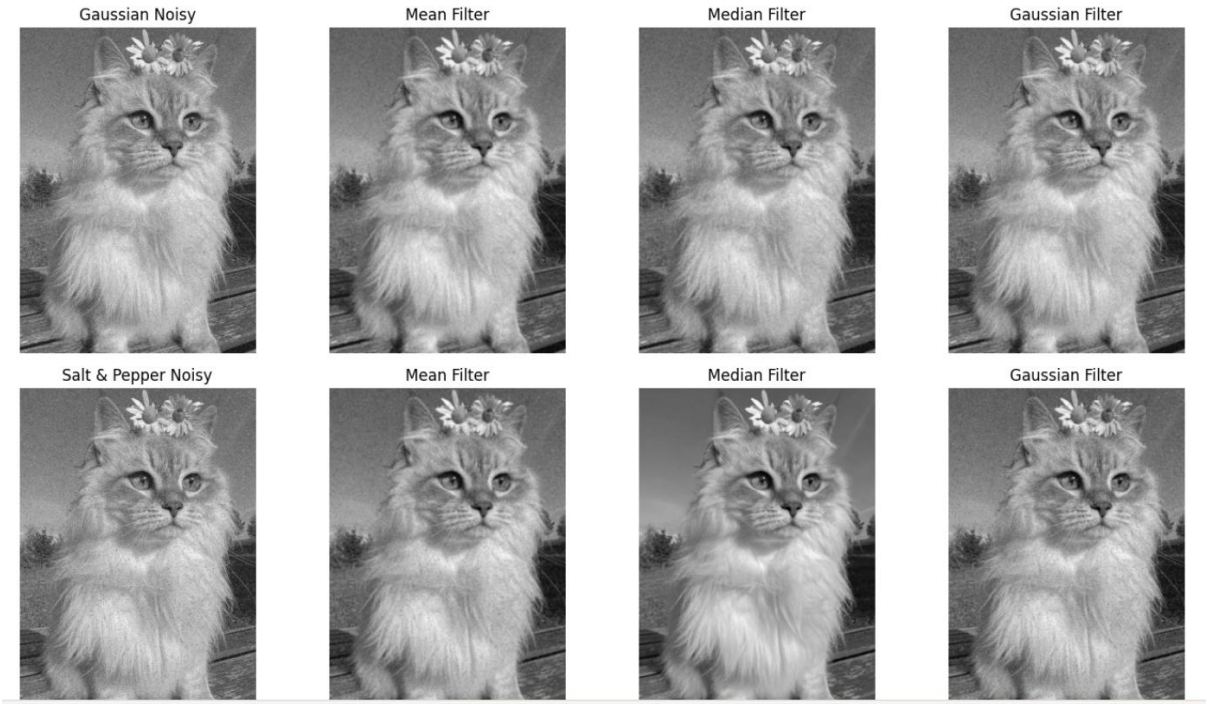


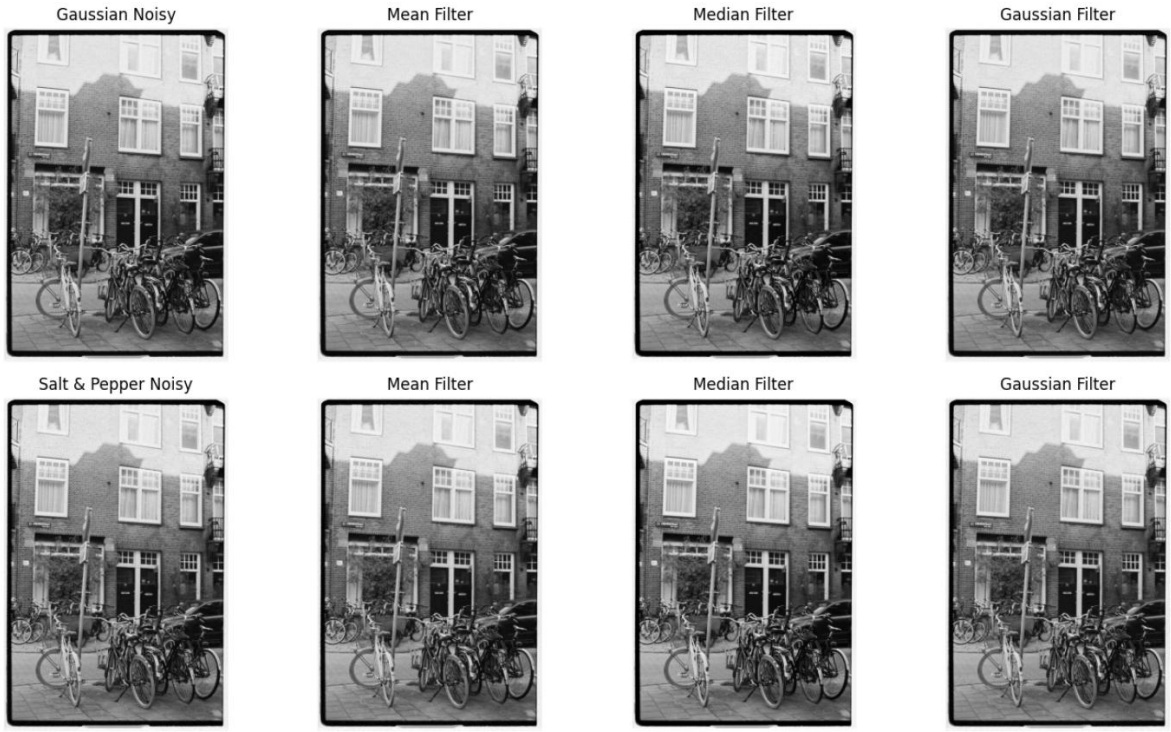
Task 3: Image Restoration Techniques

```
import cv2

import matplotlib.pyplot as plt


mean_gaussian = cv2.blur(gaussian_noisy, (5,5))

median_gaussian = cv2.medianBlur(gaussian_noisy, 5)

gaussian_gaussian = cv2.GaussianBlur(gaussian_noisy, (5,5), 0)


mean_sp = cv2.blur(sp_noisy, (5,5))

median_sp = cv2.medianBlur(sp_noisy, 5)

gaussian_sp = cv2.GaussianBlur(sp_noisy, (5,5), 0)


plt.figure(figsize=(14,8))


plt.subplot(2,4,1)

plt.title("Gaussian Noisy")

plt.imshow(gaussian_noisy, cmap='gray')
```

```
plt.axis('off')

plt.subplot(2,4,2)
plt.title("Mean Filter")
plt.imshow(mean_gaussian, cmap='gray')
plt.axis('off')

plt.subplot(2,4,3)
plt.title("Median Filter")
plt.imshow(median_gaussian, cmap='gray')
plt.axis('off')

plt.subplot(2,4,4)
plt.title("Gaussian Filter")
plt.imshow(gaussian_gaussian, cmap='gray')
plt.axis('off')

plt.subplot(2,4,5)
plt.title("Salt & Pepper Noisy")
plt.imshow(sp_noisy, cmap='gray')
plt.axis('off')

plt.subplot(2,4,6)
plt.title("Mean Filter")
plt.imshow(mean_sp, cmap='gray')
plt.axis('off')

plt.subplot(2,4,7)
```

```
plt.title("Median Filter")
plt.imshow(median_sp, cmap='gray')
plt.axis('off')

plt.subplot(2,4,8)
plt.title("Gaussian Filter")
plt.imshow(gaussian_sp, cmap='gray')
plt.axis('off')

plt.tight_layout()
plt.show()
```





Task 4: Performance Evaluation

```
mean_gaussian = cv2.blur(gaussian_noisy, (5,5))

median_gaussian = cv2.medianBlur(gaussian_noisy, 5)

gaussian_gaussian = cv2.GaussianBlur(gaussian_noisy, (5,5), 0)


mean_sp = cv2.blur(sp_noisy, (5,5))

median_sp = cv2.medianBlur(sp_noisy, 5)

gaussian_sp = cv2.GaussianBlur(sp_noisy, (5,5), 0)


plt.figure(figsize=(14,8))


plt.subplot(2,4,1)

plt.title("Gaussian Noisy")

plt.imshow(gaussian_noisy, cmap='gray')
```

```
plt.axis('off')

plt.subplot(2,4,2)
plt.title("Mean Filter")
plt.imshow(mean_gaussian, cmap='gray')
plt.axis('off')

plt.subplot(2,4,3)
plt.title("Median Filter")
plt.imshow(median_gaussian, cmap='gray')
plt.axis('off')

plt.subplot(2,4,4)
plt.title("Gaussian Filter")
plt.imshow(gaussian_gaussian, cmap='gray')
plt.axis('off')

plt.subplot(2,4,5)
plt.title("Salt & Pepper Noisy")
plt.imshow(sp_noisy, cmap='gray')
plt.axis('off')

plt.subplot(2,4,6)
plt.title("Mean Filter")
plt.imshow(mean_sp, cmap='gray')
plt.axis('off')

plt.subplot(2,4,7)
```

```
plt.title("Median Filter")
plt.imshow(median_sp, cmap='gray')
plt.axis('off')

plt.subplot(2,4,8)
plt.title("Gaussian Filter")
plt.imshow(gaussian_sp, cmap='gray')
plt.axis('off')

plt.tight_layout()
plt.show()
```

	Filter	MSE	PSNR
0	Mean (Gaussian Noise)	61.779962	30.222327
1	Median (Gaussian Noise)	65.758256	29.951301
2	Gaussian (Gaussian Noise)	63.273166	30.118608
3	Mean (Salt & Pepper)	62.252800	30.189215
4	Median (Salt & Pepper)	48.851316	31.242041
5	Gaussian (Salt & Pepper)	58.368538	30.469015

	Filter	MSE	PSNR
0	Mean (Gaussian Noise)	56.385652	30.619118
1	Median (Gaussian Noise)	54.464863	30.769639
2	Gaussian (Gaussian Noise)	56.455847	30.613714
3	Mean (Salt & Pepper)	55.652870	30.675928
4	Median (Salt & Pepper)	32.386060	33.027222
5	Gaussian (Salt & Pepper)	44.832913	31.614834

Task 5: Analytical Discussion

```
best_gaussian = df.iloc[0:3].loc[df.iloc[0:3]["PSNR"].idxmax()]
best_sp = df.iloc[3:6].loc[df.iloc[3:6]["PSNR"].idxmax()]
```

```
print("\nPerformance Comparison:")
print(df)

print("\nBest Filter for Gaussian Noise:")
print(best_gaussian)

print("\nBest Filter for Salt & Pepper Noise:")
print(best_sp)
```

```
Performance Comparison:
                      Filter        MSE       PSNR
0      Mean (Gaussian Noise)  61.779962  30.222327
1    Median (Gaussian Noise)  65.758256  29.951301
2  Gaussian (Gaussian Noise)  63.273166  30.118608
3       Mean (Salt & Pepper)  62.252800  30.189215
4     Median (Salt & Pepper)  48.851316  31.242041
5   Gaussian (Salt & Pepper)  58.368538  30.469015

Best Filter for Gaussian Noise:
Filter    Mean (Gaussian Noise)
MSE                   61.779962
PSNR                  30.222327
Name: 0, dtype: object

Best Filter for Salt & Pepper Noise:
Filter    Median (Salt & Pepper)
MSE                    48.851316
PSNR                   31.242041
Name: 4, dtype: object
```

```
Performance Comparison:
                      Filter        MSE       PSNR
0      Mean (Gaussian Noise)  56.385652  30.619118
1    Median (Gaussian Noise)  54.464863  30.769639
2  Gaussian (Gaussian Noise)  56.455847  30.613714
3       Mean (Salt & Pepper)  55.652870  30.675928
4     Median (Salt & Pepper)  32.386060  33.027222
5   Gaussian (Salt & Pepper)  44.832913  31.614834

Best Filter for Gaussian Noise:
Filter    Median (Gaussian Noise)
MSE                     54.464863
PSNR                    30.769639
Name: 1, dtype: object

Best Filter for Salt & Pepper Noise:
Filter    Median (Salt & Pepper)
MSE                     32.38606
PSNR                   33.027222
Name: 4, dtype: object
```